

CSCE 775: Deep Reinforcement Learning Homework #1

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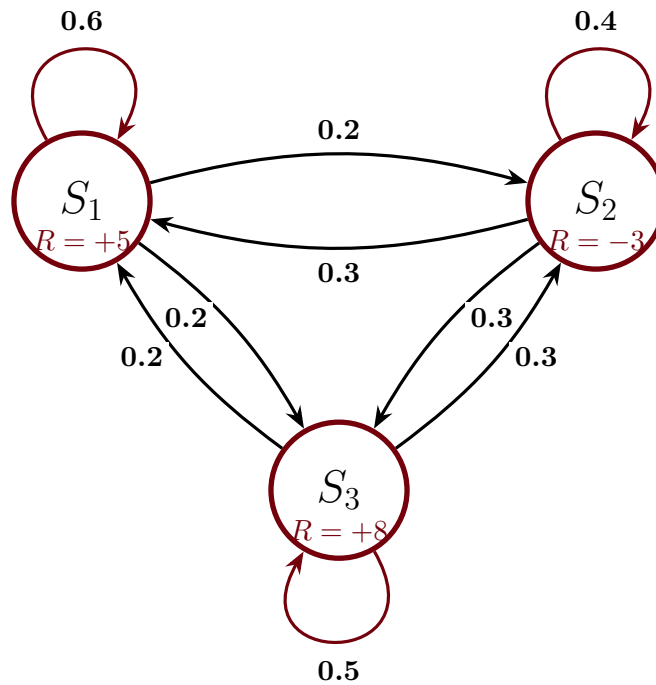
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Instructions: Show all work. Write clearly and box final answers. You may use a calculator for arithmetic but all reasoning must be shown by hand.

Problem 1: MRP State Values

Consider the following Markov Reward Process with three states $\mathcal{S} = \{S_1, S_2, S_3\}$.



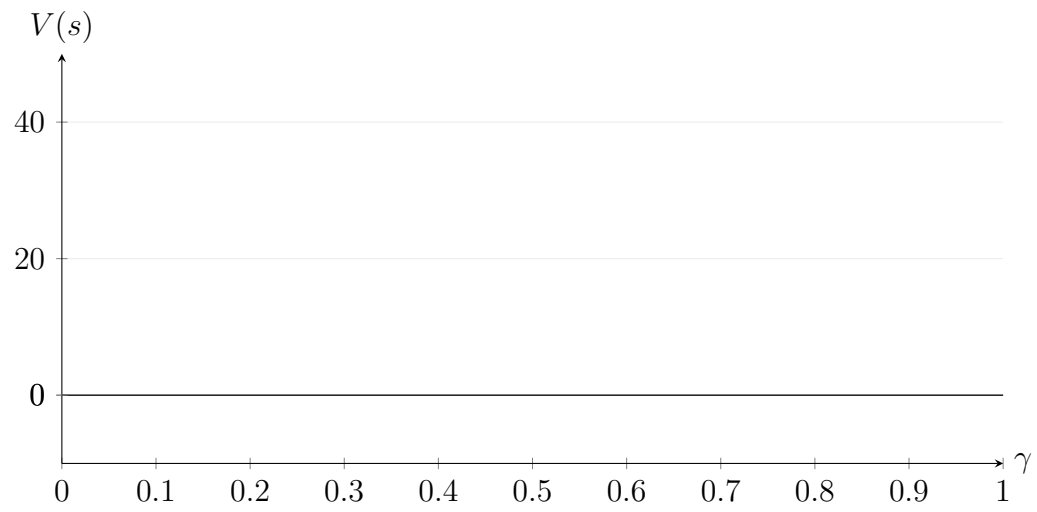
The transition matrix and reward vector are:

$$\mathcal{P} = \begin{pmatrix} 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} +5 \\ -3 \\ +8 \end{pmatrix}$$

- (a) Write the Bellman equation for this MRP in matrix form for a general discount factor γ . That is, express $\mathbf{V}$ as a function of $\mathbf{R}$, γ , and $\mathcal{P}$.

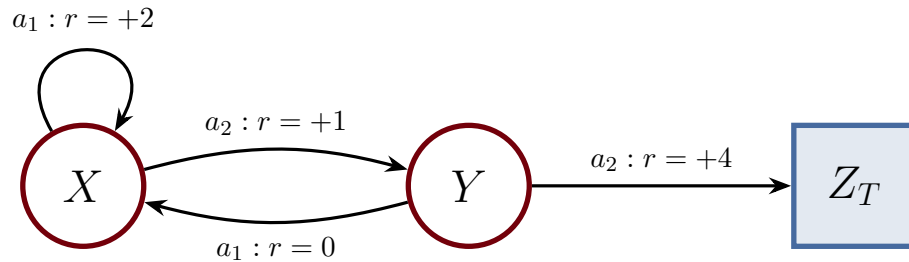
- (b) Form the matrix $(\mathbf{I} - \gamma\mathcal{P})$ explicitly for $\gamma = 0.8$. Show all entries.

- (c) Solve the 3×3 linear system $(\mathbf{I} - \gamma\mathcal{P})\mathbf{V} = \mathbf{R}$ for $\gamma = 0.8$ to find $V(S_1)$, $V(S_2)$, and $V(S_3)$. Show your work (Gaussian elimination, Cramer's rule, or equivalent).



Problem 3: Value Iteration by Hand

Consider the following MDP with states $\{X, Y\}$ and a terminal state Z_T , discount factor $\gamma = 0.9$. Each non-terminal state has two actions: a_1 and a_2 . All transitions are deterministic.



Transition and reward summary:

State	Action	Next State	Reward
X	a_1	X	$+2$
X	a_2	Y	$+1$
Y	a_1	X	0
Y	a_2	Z_T	$+4$

- (a) Initialize $V_0(X) = V_0(Y) = 0$. Perform **3 iterations** of value iteration. Fill in the table below. Show computation for each cell.

Iter k	$V_k(X)$	$\pi_k(X)$	$V_k(Y)$	$\pi_k(Y)$
0	0	—	0	—
1				
2				
3				

- (b) After convergence, what is the optimal policy π^* ? State $\pi^*(X)$ and $\pi^*(Y)$ and justify briefly.
- (c) Compute the exact optimal values $V^*(X)$ and $V^*(Y)$ by solving the Bellman optimality equations simultaneously (not by iteration).

Problem 4: Grid World Policy

Consider the 3×3 grid MDP below. The agent can move **Up**, **Down**, **Left**, **Right**. Moves that would leave the grid keep the agent in place. All transitions are deterministic. Discount factor $\gamma = 1$.

(0, 2)	(1, 2)	Goal $r = +10$
(0, 1)	Wall	Trap $r = -5$
Start	(1, 0)	(2, 0)

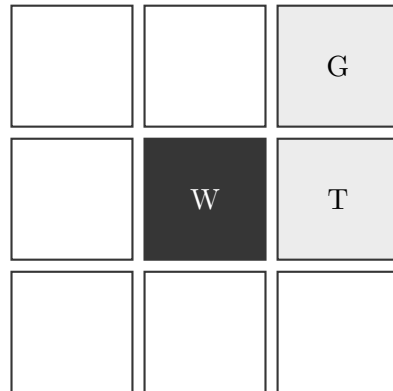
Rules:

- Every move costs $r = -1$ (step penalty), except moves into Goal or Trap.
- Reaching the **Goal** (2, 2): reward $r = +10$, episode ends.
- Reaching the **Trap** (2, 1): reward $r = -5$, episode ends.
- The **Wall** (1, 1) is impassable; moves into it keep the agent in place (still costs -1).
- Goal and Trap are terminal states.

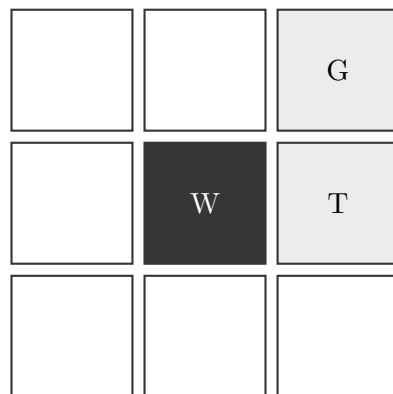
(a) List the reward $R(s, a, s')$ for every non-terminal state s and every action a . Organize your answer clearly (a table is fine).

(b) Starting from $V_0(s) = 0$ for all states, perform **2 iterations** of value iteration. Write the resulting values in each cell of the grids below.

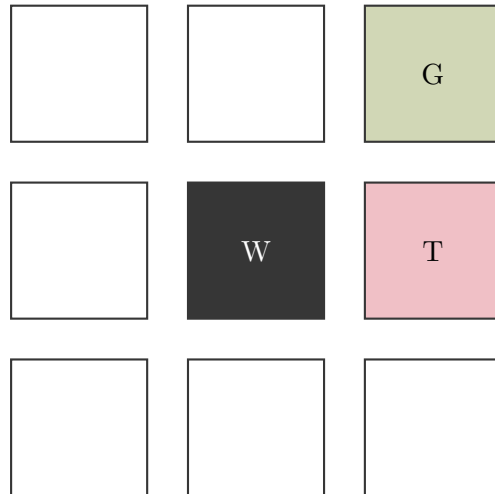
After Iteration 1:



After Iteration 2:

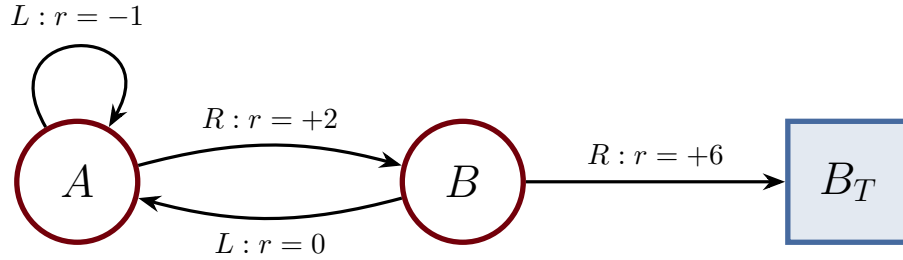


- (c) On the grid below, draw the optimal policy (one arrow per non-terminal cell indicating the best action).



Problem 5: Q-Learning Episode Trace

Consider the following 2-state MDP with states $\{A, B\}$ and terminal state B_T . Each state has two actions: L and R .



All transitions are deterministic. Use **Q-learning** with $\alpha = 0.2$, $\gamma = 1$, and $\varepsilon = 0$ (pure greedy; break ties alphabetically by action name, i.e., L before R).

Initialize all Q-values to zero: $Q(A, L) = Q(A, R) = Q(B, L) = Q(B, R) = 0$.

The three episode trajectories are given (not chosen by your policy—just apply the updates):

Episode 1: $A \xrightarrow{R} B \xrightarrow{R} B_T$

Episode 2: $A \xrightarrow{R} B \xrightarrow{R} B_T$

Episode 3: $A \xrightarrow{L} A \xrightarrow{R} B \xrightarrow{R} B_T$

(a) For each step of each episode, compute the Q-learning update. Fill in the table.

Recall: $Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$

Ep.	Step	s	a	s'	r	New $Q(s, a)$
1	1	A	R	B	+2	
1	2	B	R	B_T	+6	
2	1	A	R	B	+2	
2	2	B	R	B_T	+6	
3	1	A	L	A	-1	
3	2	A	R	B	+2	
3	3	B	R	B_T	+6	

(b) State all four Q-values after the 3 episodes are complete.

(c) What is the greedy policy $\pi(s) = \arg \max_a Q(s, a)$ for each state?

Problem 6: Exploration vs. Exploitation

- (a) Suppose an agent is in a state with three available actions $\{a_1, a_2, a_3\}$ and current Q-values:

$$Q(s, a_1) = 5.0, \quad Q(s, a_2) = 3.2, \quad Q(s, a_3) = 7.1$$

Under an ε -greedy policy with $\varepsilon = 0.4$, compute the probability of selecting each action. Show your work.

- (b) Suppose ε starts at $\varepsilon_0 = 1.0$ and decays multiplicatively each episode by a factor of 0.99:

$$\varepsilon_k = \varepsilon_0 \cdot 0.99^k$$

After how many episodes k is $\varepsilon_k < 0.05$? Solve exactly using logarithms.

- (c) In 3–5 sentences, explain: Why is exploration important in the early stages of learning? Why is it beneficial to decay ε over time rather than keeping it fixed?